

ASHUTOSH

Email: clashutosh04@gmail.com , Phone: +91 84492 27868  
Github: <https://github.com/Ashutosh-Repos>, LinkedIn: <https://www.linkedin.com/in/ashutosh-kumar-2867182a2>  
Leetcode: <https://leetcode.com/u/ashutosh0406> Portfolio: <https://portfolio-3-0-blond.vercel.app>

Technical Skills

Languages: Typescript, Javascript, Go, Java, Python, SQL  
Backend & Networking: Express.js, net/http, gRPC, HTTP 1.1/2/3, TCP/UDP, Server-Sent Events (SSE), WebSockets  
Frontend: React.js, Next.js, HTML5, CSS3, TailwindCss  
Databases & Cloud Infra: PostgreSQL, MongoDB, SQLite, Prisma, Mongoose, NATS JetStream, Redis, S3 Object Storage  
Tools & AI: Git , Docker , LLM Integration (OpenAI, Gemini, Claude)

Experience

InAmigos Foundation (Web Development Intern) , May 2026

- **Responsive UI Development:** Built responsive UI views for the event registration and volunteer signup portals.
- **Backend & Latency Optimization:** Refactored Express.js REST API routes and database query layers, reducing backend latency by 30% for high-traffic social project signup endpoints.

Projects

Tessera — Distributed Transcoding Engine (Go, Docker, Redis) (Github - [github.com/Ashutosh-Repos/Tessera](https://github.com/Ashutosh-Repos/Tessera))  
(Single line description - A distributed and scalable video transcoding system.)

- **Stateless API:** Designed a stateless Go API utilizing S3 presigned multipart URLs, keeping gateway memory footprints under 50MB RAM by routing file uploads directly from client browsers to object storage.
- **Consistent Hash Ring Routing:** Engineered a virtual-node consistent hash ring with Etcd-lease discovery to route video slicing tasks without centralized database polling or coordination lock contention.
- **Multimedia Codec Pipelines:** Integrated FFmpeg segmenters to process H.264/AVC video files into keyframe-aligned 5-second raw segment chunks, compiling final HLS master playlists and DASH MPD manifests.
- **Memory-Optimized Slicing:** Built a zero-copy stream slicer detecting MP4 moov atoms in the first 1MB, piping S3 socket streams directly into memory-based FFmpeg segments, avoiding local disk write overhead.
- **Network Socket Streams:** Built a Go edge gateway and event multiplexer utilizing TCP sockets, REST APIs, and Server-Sent Events (SSE) to fan out progress updates to clients over a single Redis stream connection.
- **Automated Integration Testing:** Developed an integration test suite that spawns local Redis, NATS, Etcd, and MinIO instances, dynamically generates test videos using FFmpeg, and verifies the full ingestion-to-manifest compilation lifecycle.

Shai - AI Powered CLI Assistant (Typescript, Node.js, React, Ink, SQLite)  
(npm - <https://www.npmjs.com/package/shai-shell>) (Github - [github.com/Ashutosh-Repos/ShAI](https://github.com/Ashutosh-Repos/ShAI))  
(Single Line Description - A command-line utility, it translates devs natural language commands to relevant shell commands.)

- **Object-Oriented Design:** Designed modular class structures in TypeScript applying clean OOP design patterns to separate user configuration files, database managers, and LLM API clients.
- **Interactive TUI:** Built interactive terminal console using React and Ink to render asynchronous AI token streams in real time.
- **Automated Validation Layer:** Programmed a security validation module that parses and inspects shell strings before system execution to intercept potentially destructive commands.
- **Unified LLM Integration:** Integrated multiple LLM (Claude, OpenAI, Gemini, Ollama) into a unified wrapper to support dynamic model selection.
- **Encrypted Offline Storage:** Built an offline prompt history engine by using local SQLite with upstream model clients.

Education

- **Madan Mohan Malaviya University of Technology, Gorakhpur** | B.Tech (Computer Science Engineering) , CGPA: 7.32
- **John Milton Public School (CBSE)** | Intermediate (XII<sup>th</sup>) , Percentage: 77.8%
- **S.M.P.K.D Shikshan Sansthan (CBSE)** | Matriculation (X<sup>th</sup>) , Percentage: 75.16%

Achievements

- NPTEL Introduction to Machine Learning (IIT Madras) | Elite Certification (Scored 73%) (Certificate Link)
- AWS Educate – Getting Started with Storage (AWS Storage Services) (Certificate Link)